

Dr. rer. nat. Xuke Hu

German Aerospace Center (DLR), Institute of Data Science

Mälzerstr. 3-5, 07745 Jena, Germany, xuke.hu@dlr.de

DLR is Germany's national research center for aeronautics and space

Personal Details

Surname: Hu

Nationality: Chinese

Birthplace: Suizhou City, China

Website: Google Scholar; ResearchGate

Forename: Xuke

Gender: Male

Birthday: December 20, 1989

Research Experience

Senior Researcher (Permanent)

September 2023 – Present

German Aerospace Center, Institute of Data Science

Jena, Germany

- Geographic information extraction from texts
- Geospatial Artificial Intelligence(GeoAI)
- Indoor mapping, localization, and navigation
- Volunteered geographic information (VGI)

Postdoc

December 2019 – August 2023

German Aerospace Center, Institute of Data Science

Jena, Germany

- Geographic information extraction from texts
- Utilizing social media data for urban planning and disaster management
- Indoor mapping, localization, and navigation
- Spatiotemporal big data analysis

Education Experience

Heidelberg University

October 2015 – October 2019

Ph.D. in Geoinformatics

Heidelberg, Germany

- Thesis: Building semantic reasoning by using rules based on available geospatial information
- First Supervisor: Prof. Dr. Alexander Zipf
- Second Supervisor: Prof. Dr. Hongchao Fan

China University of Geosciences (Wuhan)

September 2012 – July 2015

Master in Software Engineering

Wuhan, China

- Thesis: Graph Model-Aided Wi-Fi Fingerprinting
- Supervisor: Prof. Dr. Jianga Shang

China University of Geosciences (Wuhan)

September 2008 – July 2012

Bachelor in Software Engineering

Wuhan, China

Research Interests

My primary research focus lies in GeoAI. Currently, I am particularly interested in the extraction and analysis of geospatial information embedded in large volumes of unstructured and multilingual natural language text. The ultimate goal of my research is to uncover new geospatial perspectives from texts, thereby benefiting a wide range of domains, and to make such information equally accessible to researchers and practitioners across different languages and regions.

Honors & Awards

- Performance Bonus (Leistungsprämie), German Aerospace Center 2025
- Performance Bonus (Leistungsprämie), German Aerospace Center 2024
- Performance Bonus (Leistungsprämie), German Aerospace Center 2023
- Performance Bonus (Leistungsprämie), German Aerospace Center 2022
- Travel Award, ISPRS International Journal of Geo-information 2021
- Travel Grant, ISPRS Committee 2017

Professional Activities

• **Editorial service**

- Editorial Board Member, The Photogrammetric Record since 2026
- Guest Editor, International Journal of Geographical Information Science 2024
- Guest Editor, Information Processing & Management 2023
- Guest Editor, Geo-spatial Information Science 2025
- Youth Editorial Board Member, Information Geography since 2025
- Youth Editorial Board Member, Journal of Spatio-temporal Information 2025

• **Conference and community leadership**

- Chair, Joint ISPRS/ISDE Conference on Geo-digitalization for SDGs 2025
- Session Chair, ISPRS Congress 2026
- Chair, First-Fourth GeoEXT Workshops at ECIR 2023–2026
- Chair, 7th Geospatial Humanities Workshop at ACM SIGSPATIAL 2023
- Chair, Crowdsourcing for Global Mapping Workshop at ISPRS Geospatial Week 2023
- Session Chair and Panelist, APWeb-WAIM International Joint Conference on Web and Big Data 2023

• **Scientific committees and governance**

- Scientific Council Member, DLR Institute of Data Science 2025–2027
- Fellow, Royal Geographical Society since 2026
- Secretary, ISPRS ICWG (Technical Commission IV/III) 2023–2026
- Scientific Committee Member, ISPRS TC III Mid-term Symposium 2024
- Scientific Committee Member, MEDDOPLACE Shared Task 2023

Skills

Languages: Chinese (Native), English (Excellent), German (Good, B1)

Programming Languages: Java, C, C++, Python, R, Matlab

Teaching & Mentoring

Teaching at Friedrich Schiller University Jena

Since 2021

- *Geo404*: Instructed graduate students in Geography and Computer Science, co-teaching with Dr. Jens Kersten.

Ph.D. Supervision

Since 2025

- **Jan Bongard**, Ph.D. candidate (Friedrich Schiller University Jena):
"Quality-Aware and Multimodal Geospatial Data Fusion for Decision-Making."

Thesis Supervision

since 2021

- **Tim Grünemay**, B.Sc. (Friedrich Schiller University Jena): "Comparison of Geoparsing Methods for Social Media Short Messages."
- **Justin Jose**, M.Sc. (Ilmenau University of Technology): "Automated Classification of Sleep-Disruptive Audio Events Under Real World Conditions."
- **Ahmed Abdelsalam**, M.Sc. (Technische Universität München): "Estimating Building Attributes from Street-level Imagery using Vision-Language Models."

Selected Journal Publications

1. **Hu, X.**, Sun, Y., Hecking, T., Kersten, J., & Klan, F. (2025). UniTopRank: A scalable and language-independent method for toponym resolution. *International Journal of Geographical Information Science*. (Under review)
2. **Hu, X.**, Kersten, J., Klan, F., & Farzana, S. M. (2024). Toponym resolution leveraging lightweight and open-source large language models and geo-knowledge. *International Journal of Geographical Information Science*, 1-28. DOI: 10.1080/13658816.2024.2405182
3. **Hu, X.**, Elßner, T., Zheng, S., Serere, H. N., Kersten, J., Klan, F., & Qiu, Q. (2024). DLRGeoTweet: A comprehensive social media geocoding corpus featuring fine-grained places. *Information Processing & Management*, 61(4), 103742. DOI: 10.1016/j.ipm.2024.103742
4. Gu, F., Guo, F., Yu, F., Long, X., Chen, C., Liu, K., **Hu, X.**, & Guo, S. (2024). Accurate and efficient floor localization with scalable spiking graph neural networks. *Satellite Navigation*, 5(1), 6.
5. **Hu, X.**, Zhou, Z., Li, H., Hu, Y., Gu, F., Kersten, J., Fan, H., & Klan, F. (2023). Location reference recognition from texts: A survey and comparison. *ACM Computing Surveys*, 56(5), 1-37. DOI: 10.1145/3625819
6. **Hu, X.**, Sun, Y., Kersten, J., Zhou, Z., Klan, F., & Fan, H. (2023). How can voting mechanisms improve the robustness and generalizability of toponym disambiguation?. *International Journal of Applied Earth Observation and Geoinformation*, 117, 103191. DOI: 10.1016/j.jag.2023.103191
7. **Hu, X.**, Zhou, Z., Sun, Y., Kersten, J., Klan, F., Fan, H., & Wiegmann, M. (2022). GazPNE2: A general place name extractor for microblogs fusing gazetteers and pretrained transformer models. *IEEE Internet of Things Journal*, 9(17), 16259-16271. DOI: 10.1109/JIOT.2022.3150967
8. **Hu, X.**, Al-Olimat, H. S., Kersten, J., Wiegmann, M., Klan, F., Sun, Y., & Fan, H. (2022). GazPNE: annotation-free deep learning for place name extraction from microblogs leveraging gazetteer and synthetic data by rules. *International Journal of Geographical Information Science*, 36(2), 310-337.
9. **Hu, X.**, Noskov, A., Fan, H., Novack, T., Li, H., Gu, F., Shang, J., & Zipf, A. (2021). Tagging the main entrances of public buildings based on OpenStreetMap and binary imbalanced learning. *International Journal of Geographical Information Science*, 35(9), 1773-1801.
10. Sun, Y., **Hu, X.** & Xie, J., (2021). Spatial inequalities of COVID-19 mortality rate in relation to socioeconomic and environmental factors across England. *Science of The Total Environment*, 758, p.143595.
11. **Hu, X.**, Hu, X., Fan, H., Noskov, A., Wang, Z., Zipf, A., Gu, F., & Shang, J. (2021). Room semantics inference using random forest and relational graph convolutional networks: A case study of research building. *Transactions in GIS*, 25(1), 71-111.
12. **Hu, X.**, Ding, L., Shang, J., Fan, H., Novack, T., Noskov, A., & Zipf, A. (2020). Data-driven approach to learning salience models of indoor landmarks by using genetic programming. *International Journal of Digital Earth*, 13(11), 1230-1257.

13. Gu, F., **Hu, X.**, Ramezani, M., Acharya, D., Khoshelham, K., Valaee, S., & Shang, J. (2019). Indoor localization improved by spatial context—A survey. *ACM Computing Surveys (CSUR)*, 52(3), 1–35. (Corresponding author).
14. Wu, Y., Shang, J., Chen, P., Zlatanova, S., **Hu, X.**, & Zhou, Z. (2021). Indoor mapping and modeling by parsing floor plan images. *International Journal of Geographical Information Science*, 35(6), 1205–1231.
15. Sun, Y., **Hu, X.**, Huang, Y., & On Chan, T., (2020). Spatial Patterns of Childhood Obesity Prevalence in Relation to Socioeconomic Factors across England. *ISPRS International Journal of Geo-Information*, 9(10), p.599.
16. Sun, Y., Wang, S., Xie, J., & **Hu, X.**, (2020). Modeling Local-Scale Violent Crime Rate: A Comparison of Eigenvector Spatial Filtering Models and Conventional Spatial Regression Models. *The Professional Geographer*, pp.1–10.
17. **Hu, X.**, Fan, H., Noskov, A., Zipf, A., Wang, Z., & Shang, J. (2019). Feasibility of using grammars to infer room semantics. *Remote Sensing*, 11(13), 1535.
18. **Hu, X.**, Fan, H., & Noskov, A. (2017). Roof model recommendation for complex buildings based on combination rules and symmetry features in footprints. *International Journal of Digital Earth*. 11(10), pp.1039–1063.
19. Sun, Y., Mobasher, A., **Hu, X.**, & Wang, W., (2017). Investigating impacts of environmental factors on the cycling behavior of bicycle-sharing users. *Sustainability*, 9(6), p.1060.
20. Shang, J., **Hu, X.**, Cheng, W., & Fan, H. (2016). GridiLoc: A backtracking grid filter for fusing the grid model with PDR using smartphone sensors. *Sensors*, 16(12), 2137. (Corresponding author)
21. **Hu, X.**, Shang, J., Gu, F., & Han, Q. (2015). Improving Wi-Fi indoor positioning via AP sets similarity and semi-supervised affinity propagation clustering. *International Journal of Distributed Sensor Networks*, 11(1), 109642.
22. Shang, J., **Hu, X.**, Gu, F., Wang, D. & Yu, S., 2015. Improvement schemes for indoor mobile location estimation: A survey. *Mathematical Problems in Engineering*, 2015.

Conference Papers

1. **Hu, X.**, Kersten, J., & Klan, F. (2025). Scalable Toponym Resolution with LLMs: Accuracy and Speed Optimizations. Third International Workshop on Geographic Information Extraction from Texts (GeoExT) at 47th ECIR.
2. **Hu, X.**, & Kersten, J. (2024). Enhancing Toponym Resolution with Fine-Tuned LLMs (Llama2). Second International Workshop on Geographic Information Extraction from Texts (GeoExT) at 46th ECIR.
3. **Hu, X.**, Sun Y., Kersten, J., & Klan, F. (2022). How can voting mechanisms improve the robustness of individual toponym resolution approaches? LBS2022.
4. Gu, F., Sng, W., **Hu, X.**, & Yu, F. (2021). EventDrop: data augmentation for event-based learning. The 30th International Joint Conference on Artificial Intelligence (IJCAI).
5. **Hu, X.**, Zhou Z., Kersten, J., & Klan, F. (2021). A general and robust location extraction approach by fusing gazetteers and transformer models. 11th International Conference on Geographic Information Science. DOI: 10.4230/LIPIcs.GIScience.2021.53
6. **Hu, X.**, Fan, H., Zipf, A., Shang, J., & Gu, F. (2017). A conceptual framework for indoor mapping by using Grammars. *ISPRS Geospatial Week 2017*, 18–22 September, Wuhan, China.

Patents

Floor Identification Method and System Based on Graph Pulse Neural Network

- Invention Patent | CN116186581A | 2023 | Granted by CNIPA
- Inventors: Fuqiang Gu, Fangming Guo, Fangwen Yu, Chao Chen, Kai Liu, Songtao Guo, **Xuke Hu**, Jianga Shang

An Indoor Navigation Landmark Extraction Method

- Invention Patent | CN110569323A | 2022 | Granted by CNIPA
- Inventors: Jianga Shang, **Xuke Hu**, Lei Ding

Discrete Asynchronous Event Data Enhancement Method

- Invention Patent | CN113177640B | 2022 | Granted by CNIPA
- Inventors: Fuqiang Gu, Fangwen Yu, **Xuke Hu**

Grid Model - Based PDR Indoor Positioning Method and System

- Invention Patent | CN105606102B | 2017 | Granted by CNIPA
- Inventors: Jianga Shang, **Xuke Hu**, Wen Cheng

Indoor Emergency Response Context - Aware Navigation System and Method

- Invention Patent | CN103913165B | 2016 | Granted by CNIPA
- Inventors: Jianga Shang, Fangwen Yu, **Xuke Hu**, Zhiyong Zhou, Jinjin Yan, Xiao Zhang, Xiaonan Wang, Xinyi Tang

Wi-Fi Fingerprint Positioning Method Based on AP Collection Similarity

- Invention Patent | CN103476115B | 2016 | Granted by CNIPA
- Inventors: Jianga Shang, **Xuke Hu**, Fangwen Yu, Jinjin Yan, Fuqiang Gu

Developed Open-Source Software & Datasets

GazPNE	2021
Deep learning-based place name extractor using gazetteers.	GitHub: GazPNE
GazPNE2	2022
Place name extractor leveraging BERT, BERTweet, and gazetteers.	GitHub: GazPNE2
Voting	2023
Voting-based approach for toponym resolution.	GitHub: Voting
LLM-Geocoding	2024 – Present
Lightweight LLM-based toponym resolution.	GitHub: LLM-Geocoding
DLRGeoTweet	2024 – Present
Geocoding dataset with extensive fine-grained places.	GitHub: DLRGeoTweet